

WISDOM BENSON

COMPUTATIONAL MATERIALS PHYSICIST | SCIENTIFIC WORKFLOW BUILDER

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PROFESSIONAL PROFILE

Physics PhD researcher at Howard University integrating first-principles simulation, Raman spectroscopy, high-performance computing, and reproducible scientific workflows to investigate ZnO quantum dots and mixed Sn-Pb perovskites. First-author researcher with publications in AIP Advances and the Journal of Physics and Chemistry of Solids.

TECHNICAL FOCUS

Electronic structure: DFT, DFPT, PDEP-GW, many-body perturbation theory | **Platforms:** Quantum ESPRESSO, WEST, SIESTA, SDSC Expanse, ANL-CNM | **Characterization:** Raman and optical spectroscopy, thin-film synthesis | **Computing:** Python, C++, Java, MATLAB, LaTeX

RESEARCH AND TEACHING EXPERIENCE

Graduate Research & Teaching Assistant | Howard University

Aug. 2024 - Present

- Conduct first-principles DFT and PDEP-GW calculations with Quantum ESPRESSO and WEST on SDSC Expanse and ANL-CNM resources, including convergence, restart, and validation workflows.
- Model electronic structure, phonon behavior, passivation chemistry, frontier levels, and piezoelectric response in finite ZnO nanocrystals.
- Led the summer 2025 REU research program on modeling and simulation of piezoelectric quantum dots in the Laser Spectroscopy Lab.

Graduate Research & Teaching Assistant | North Carolina Central University

Aug. 2022 - May 2024

- Synthesized and optically characterized mixed tin-lead perovskite thin films and analyzed composition-dependent experimental trends.
- Taught practical physics laboratory sections to more than 200 undergraduate students and supported academic mentoring.
- Presented research at national conferences and contributed to first-author and peer-reviewed publications.

Classroom Teacher, Physics | God's Will Academy

Feb. 2022 - July 2022

- Developed and delivered physics instruction for more than 100 secondary-school students and coordinated academic support with families and counselors.

EDUCATION

Howard University

Washington, DC | Aug. 2024 - May 2029

PhD in Physics | Computational materials and quantum-dot research

anticipated

North Carolina Central University

Durham, NC | May 2024

MS in Physics with Honors | Thesis on mixed Sn-Pb perovskite thin films

University of Nigeria, Nsukka

Nsukka, Nigeria | June 2021

BS in Physics with Honors | DFT study of lead-halide perovskites

METHODS AND TOOLS

Electronic structure: Quantum ESPRESSO, WEST/PDEP-GW, SIESTA, DFT, DFPT, many-body perturbation theory, VESTA

Scientific computing: Python, C++, Java, MATLAB, LaTeX, reproducible analysis, convergence testing

Characterization: Raman spectroscopy, optical spectroscopy, thin-film synthesis, spectral analysis

HPC workflow practice: SDSC Expanse, ANL-CNM resources, job recovery, validation, and research workflow design

RECOGNITION AND SELECTED EVIDENCE

Leadership: American Physical Society Student Ambassador, 2024-2025 and 2025-2026; NCCU International Student Award, 2024.

First author: Electronic properties of zinc oxide quantum dot. *AIP Advances* 16(2), 025236 (2026). DOI

First author: Enhanced stability and near-IR tunability in tin-lead perovskites. *Journal of Physics and Chemistry of Solids* 211, 113511 (2026). DOI

First author: Advanced computational studies of quantum dots. *Nanoelectronics*, CRC Press, pp. 169-197 (2025). DOI | Full record